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RESEARCH ARTICLE

Fully Distributed Low-Complexity UAV Formation Control With Fixed-Time Full-State Error Performance

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ABSTRACT This paper presents a fully distributed, low-complexity UAV formation controller design with fixed-time full-state error performance, which is able to address the difficulties in obtaining global information and suppressing external disturbances without observer or adaptive law in existing solutions. The position of neighboring UAVs is the only information to be exchanged, and desired state observer is not required, thereby reducing communication costs, which is well-suited to larger-scale UAV formation systems. The proposed control scheme has the capability to handle external disturbances without observer or adaptive law, simplifying the controller's design and implementation process, reducing computational burden. Moreover, by imposing fixed-time performance constraints on tracking error and virtual error, fixed-time formation tracking error and virtual velocity tracking error can be achieved, which also improves the response speed and accuracy of UAV formation, making it more capable of meeting the time requirements of actual tasks and adapting to dynamically changing environments. Additionally, all closed-loop signals are rigorously proven to be semi-globally ultimately uniformly bounded. Finally, numerical simulation is conducted to verify the effectiveness of proposed control scheme.

INDEX TERMS Fully distributed controller, UAV formation, low complexity, fixed-time preset performance.

I. INTRODUCTION

In recent years, drone swarms have demonstrated broad application prospects in various fields due to their flexibility, low cost, and environmental adaptability [1], [2], [3], [4]. However, facing dynamic and complex environments and task requirements, the key issue that remains to be researched is the formation of a swarm that can handle complex dynamic environments, and ensure efficient and stable communication [5], [6], [7].

Due to the nonlinear dynamics, insufficient driving force, strong coupling, and sensitivity to interference of multi-drone systems, the control of drone formations becomes highly challenging under uncertain flight conditions. Early control methods for drone formations based on graph theory and leader-follower models can intuitively model interactions between drones, flexibly structure topologies, and analyze systems using graph theory tools, offering flexibility and effectiveness [8], [9], [10]. Fuzzy control and reinforcement learning algorithms provide powerful control solutions for drone formations, adapting to nonlinear and uncertain environments, capable of self-learning and real-time decision-making, and functioning effectively without precise models [11], [12], [13], [14], [15], [16]. Recursive optimization methods, with their real-time dynamic adjustments and efficient computations; event-driven control, with its reduced computational burden and efficient response; and port-Hamiltonian system methods, with their advantages in energy management and stability design, collectively enhance the performance, robustness, and adaptability of drone formation systems [17], [18], [19].

However, the aforementioned control methods struggle with dynamic and complex environments. Most distributed results face difficulties in obtaining global information when

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the network scale is large and depend heavily on global information, making the stability of multi-drone formations an issue [20]. Current research in this area aims to reduce communication and control update frequency among drones to achieve tracking synchronization and simplify design complexity. For instance, an event-triggered multi-aircraft formation target tracking control algorithm was proposed to address the frequent information exchange and control update issues during multi-drone formations [21]. Another study explored bandwidth-limited orderly communication in multi-drone formations, introducing dynamic spectrum interaction and reinforcement learning algorithms, proposing two drone formation communication methods: information sharing and dynamic time slot allocation [22]. A distributed formation containment protocol based on adjacent information was designed to achieve the expected formation of the leader's state and ensure the followers' states converge to the convex hull traversed by the leader while maintaining the leader's formation as a convex combination [23]. A novel relay algorithm using Unscented Kalman Filter and Hybrid Particle Swarm Optimization (UKF-HPSO) was proposed to optimize communication throughput in drone mobile relay formations, considering challenges such as obstacle avoidance, channel complexity, high drone dynamics, and real-time task requirements, effectively optimizing system throughput through real-time prediction and optimal deployment to address unique challenges in drone communication in dynamic environments [24]. Although these methods effectively reduce communication burdens, their complexity remains high, and they cannot effectively handle external disturbances.

Regarding the issue of multiple drones experiencing nonlinear time-varying disturbances during coordinated formation flight, there is still insufficient research on the synchronization problem under disturbances [25], [26]. Although some formation control algorithms are specifically designed for multi-drone systems [10], [27], they cannot achieve disturbance suppression. External disturbances reduce system performance and cause instability. Current anti-disturbance approaches include active and passive disturbance rejection. Active disturbance rejection is more suitable for applications requiring high disturbance response and significant environmental changes [28], [29], [30], while passive disturbance rejection suits scenarios with relatively low disturbance response requirements and minor environmental changes [31]. To address disturbance rejection, a method combining a distributed fixed-time sliding mode estimator and fixed-time disturbance observer with a fixed-time sliding mode controller was proposed, enabling each drone to quickly estimate desired position information and generate disturbance estimates to counteract compound disturbances, achieving stable tracking of the desired trajectory [32]. Another study proposed a robust control method based on disturbance observers, utilizing disturbance observers to estimate unknown external disturbances and combining these estimates with the controller to enhance system stability and performance [33]. A finite-time disturbance observer was proposed to handle unknown external disturbances, and a non-singular terminal sliding mode control method was designed based on the observer's output to avoid singularity issues faced by traditional terminal sliding mode control [30]. However, these methods cannot perfectly achieve disturbance rejection. The established control schemes lack robustness to uncertainties, requiring specific adaptive parameters and additional observers for neighbor state estimation. Thus, further research is needed to enhance the control effect and robustness of quadrotor drone formation systems under unknown external disturbances.

Besides the above issues, the convergence rate is crucial for real-time formation systems. Asymptotic formation control algorithm could only achieve exponential convergence [34], [35]. Such control methods with infinite settling time are not suitable for urgent formation task. To obtain rapid convergence rate, finite-time control scheme is utilized for formation control [36]. The problem of this method is that the convergence time is depend on initial states of system. Thus, fixed-time control method, proposed by [37], is employed for controller design of formation to ensure settling time independently on initial condition [38]. However, for systems with uncertainties and disturbances, finite-time control may have weak disturbance suppression effects, and if initial conditions are large enough, the convergence time may increase significantly [39]. Fixed-time control effectively addresses this issue, as it requires no online parameter adjustment [40], [41], providing a constant upper limit on convergence time and offering more reliable control performance.

To address the above issues, we propose a new distributed performance-guaranteed control scheme to enhance system robustness to uncertainties and offer more relaxed controll ability conditions compared to the most advanced existing methods. Additionally, the scheme ensures that the output tracking error of each agent possesses arbitrarily preassigned convergence rates and features lower structural and computational complexity since it completely avoids virtual controllers and desired trajectory derivatives. Unlike the latest methods, it does not use additional observers to estimate neighbors' states, and each agent's control design only requires its own state and its neighbors' output information, ensuring that all internal signals are semi-globally ultimately uniformly bounded.

In summary, balancing control performance, convergence speed, communication efficiency, disturbance rejection, and system complexity is necessary. Therefore, a comprehensive and efficient controller design are still needed. Based on previous research, this paper aims to address the above issues by proposing a fully distributed, low-complexity fixed-time preset performance controller for drone formations. The main contributions of this work are summarized as follows:

(1) Compared with [20], [23], only the position information of neighbor is required to be transmitted for communication, solving the difficulty of obtaining global information without extra state observers. This has reduced communication costs, suitable for larger-scale UAV formation systems.

(2) Different from previous research [25], [26], [41], [42], external disturbances have been effectively managed without the need for adaptive parameters, observers, or adaptive laws. This approach simplifies the design and implementation process of the controller, reduces computational load, and enhances the real-time performance and stability of the system.
(3) By imposing fixed-time performance constraints on the tracking error and virtual error, we have achieved fixed-time formation tracking error and virtual speed tracking error. Compared to the control methods presented in [40], [41], error performance of the formation control system, capability of handling external disturbances, model errors, and variations of system parameters are improved.

The discussion will proceed as follows: Section II describes the problem formulation and prepares the introduction of UAV formation dynamics models and control objectives. Section III designs the controller and conducts stability analysis. Section IV presents simulations and demonstrations of the algorithm. Conclusions are drawn in Section V.

II. PROBLEM FORMULATION AND PRELIMINARIES

A. DYNAMICS OF THE UAV FORMATION SYSTEM

In this brief, a multi-rotor UAV system consisting of N four-rotor UAVs and a virtual pilot is studied. The internal and external loop control architecture is adopted to solve the formation control problem.

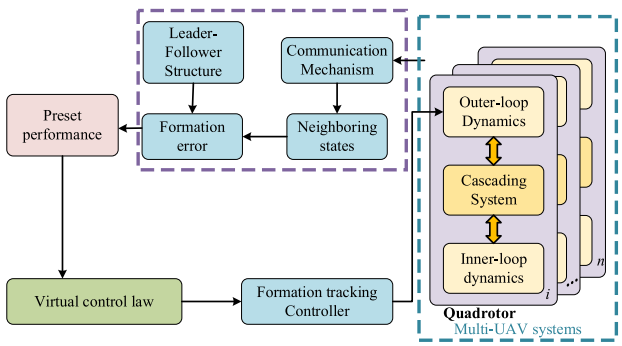

FIGURE 1. Control diagram of multi-UAV systems.

Outer loop: The position and velocity tracking controllers are designed to make the quadcopters follow the desired trajectory and maintain the prescribed formation pattern. The outer loop is designed to achieve predetermined performance, enable formation tracking, and handle the complexities of multi-quadrotor dynamics such as nonlinearity, coupling, underdrive, and external interference. The outer loop controller is responsible for tracking the predetermined motion trajectory and maintaining the predetermined formation. Inner loop: This handles the attitude stability control to stabilize the roll, pitch and yaw angles of the individual quadrotor. The inner loop controller is responsible for stabilizing the attitude angle.

The following equation represents the outer loop dynamics of the quadrotor subjected to external aerodynamic forces and disturbances:

$$\begin{aligned}\ddot{x}_i &= -\frac{P_1}{m_i}\dot{x}_i + \frac{A_i}{m_i}\cos\phi_i\sin\theta_i\cos\psi_i + \frac{A_i}{m_i}\sin\phi_i\sin\psi_i + d_{1i}\\ \ddot{y}_i &= -\frac{P_2}{m_i}\dot{y}_i + \frac{A_i}{m_i}\cos\phi_i\sin\theta_i\sin\psi_i - \frac{A_i}{m_i}\sin\phi_i\cos\psi_i + d_{2i}\\ \ddot{z}_i &= -\frac{P_3}{m_i}\dot{z}_i + \frac{A_i}{m_i}\cos\phi_i\cos\theta_i - g + d_{3i}\end{aligned} \tag{1}$$

where m_i is the mass of the quadrotor, g is the acceleration of the gravity, and A_i denotes the thrust, $P_j, j = 1, 2, 3$ is the aerodynamic damping coefficient, ϕ_i, θ_i and ψ_i represents the roll, pitch and yaw angle of the i-th quadrotor, respectively, $\boldsymbol{d}$ is the external disturbances.

To facilitate controller design of outer loop, we define the following virtual control input:

$$\begin{aligned}u_{xi} &= \frac{A_i}{m_i}\cos\phi_i\sin\theta_i\cos\psi_i + \frac{A_i}{m_i}\sin\phi_i\sin\psi_i\\ u_{yi} &= \frac{A_i}{m_i}\cos\phi_i\sin\theta_i\sin\psi_i - \frac{A_i}{m_i}\sin\phi_i\cos\psi_i\\ u_{zi} &= \frac{A_i}{m_i}(\cos\phi_i\cos\theta_i) - g\end{aligned} \tag{2}$$

Then, the kinematics and dynamics of i-th follower is given by

$$\begin{aligned}\dot{\boldsymbol{x}}_i &= \boldsymbol{v}_i\\ \dot{\boldsymbol{v}}_i &= \boldsymbol{C}_i\boldsymbol{v}_i + \boldsymbol{u}_i + \boldsymbol{d}_i\end{aligned} \tag{3}$$

B. GRAPH THEORY

In drone formation, graph theory is widely used to model and analyze the communication and relationships between drones. We can use a directed graph $\mathcal{J} = (\mathcal{Q}, \mathcal{R})$ to represent a drone formation, where $\mathcal{Q} = \{q_1, q_2, \cdots, q_n\}$ is the set of nodes, each node represents a drone, $\mathcal{R} \subseteq \mathcal{Q} \times \mathcal{Q}$ is the set of edges, and each edge represents the communication link between drones. The adjacency matrix A_{ij} is used to describe the connection relationship of a graph, where $A_{ij} = 1$ indicates the existence of a communication link between node j and node i, otherwise $A_{ij} = 0$. For a formation containing n drones, the size of its adjacency matrix $\boldsymbol{A}$ is $n \times n$.

The Laplace matrix L is another important matrix that can be obtained from the adjacency matrix $\boldsymbol{A}$ and the degree matrix $\boldsymbol{D}$. The degree matrix $\boldsymbol{D}$ is a diagonal matrix, given by

$$D_{ii} = \sum_{j=1}^{n} A_{ij} \tag{4}$$

$\boldsymbol{D}_{ii}$ indicates the degree of egress of node i, which is the number of edges connected to node i.The Laplace matrix is defined as $L = D - A$. The eigenvalues and eigenvectors of L can be used to analyze the connectivity and stability of the system. Specifically, for a formation composed of n drones, its state can be expressed as a vector $x \in R_n$, where represents the state of the i-th drone (such as position or

velocity). Let $\bar{\mathcal{J}} = \left(\bar{\mathcal{Q}}, \bar{\mathcal{R}}\right)$ be the augmented graph with $\bar{\mathcal{Q}} = \{q_0, q_1, \cdots, q_n\}$ and $\bar{\mathcal{R}} \subseteq \bar{\mathcal{Q}} \times \bar{\mathcal{Q}}$.

C. CONTROL OBJECTIVE

Let $\boldsymbol{x}_d$ be the leader, the formation tracking error of i-th follower can be described as

$$\boldsymbol{e}_i = \boldsymbol{x}_i - \boldsymbol{x}_d - \boldsymbol{\delta}_i \tag{5}$$

where $\boldsymbol{\delta}_i$ is the relative position between i-th follower and the virtual leader. In this paper, we devote to design a designing a low-complexity distributed controller such that

1) The tracking error can be evolved in the prescribed region, i.e., $\|\boldsymbol{e}_i\| \leq \rho(t)$, where $\rho(t)$ is a fixed-time prescribed performance function, satisfying

$$\rho(t) = \begin{cases} \dfrac{(\rho_0 - \rho_\infty)}{T^2}(t-T)^2 + \rho_\infty & t \leq T \\ \rho_\infty & t > T \end{cases} \tag{6}$$

Especially, every follower can only obtain position information of its neighbors and only part of followers can obtain the leader's position information, which reduces communication burden.

2) With the existence of external disturbances, all signals of closed-loop distributed system are semi-globally ultimately uniformly bounded. Especially, the adaptive parameters are not necessary.

3) Both of tracking error and virtual velocity error are restricted in fixed-time prescribed boundary.

Assumption 1: A spanning tree is contained in the augmented graph $\bar{\mathcal{J}}$ with the leader served as the root.

Assumption 2: Part of followers are able to obtain the desired trajectory $\boldsymbol{x}_d$ and $\dot{\boldsymbol{x}}_d$ is bounded.

Assumption 3: The uncertainty matrices C_i and external disturbances $\boldsymbol{d}_i$ are bounded.

III. CONTROLLER DESIGN

A. PRESCRIBED PERFORMANCE TRANSFORMATION FUNCTION

To alleviate the communication burden, we assume that every follower can only obtain position information of its neighbors and only part of followers can obtain the leader's position information. Hence, the tracking error $\boldsymbol{e}_i = \left[e_{i,1}, e_{i,2}, e_{i,3}\right]^T$ defined in (5) cannot be utilized for controller design. To this end, let $\boldsymbol{\delta}_{i,k} = \boldsymbol{\delta}_i - \boldsymbol{\delta}_k$, then auxiliary tracking error $\widehat{\boldsymbol{e}}_i$ is utilized for controller design, given by

$$\widehat{\boldsymbol{e}}_i = \sum_{k \in \mathcal{N}} a_{i,k}\left(\boldsymbol{x}_i - \boldsymbol{x}_k - \boldsymbol{\delta}_{i,k}\right) + b_i\left(\boldsymbol{x}_i - \boldsymbol{x}_d - \boldsymbol{\delta}_i\right) \tag{7}$$

In view of (5) and (7), it follows that

$$\widehat{\boldsymbol{e}}(t) = \left((\boldsymbol{L} + \boldsymbol{B}) \otimes \boldsymbol{E}_3\right)\boldsymbol{e}(t) \tag{8}$$

where $\boldsymbol{e} = \left[\boldsymbol{e}_1, \boldsymbol{e}_1, \cdots, \boldsymbol{e}_N\right]^T$, $\widehat{\boldsymbol{e}} = \left[\widehat{\boldsymbol{e}}_1, \widehat{\boldsymbol{e}}_2, \cdots, \widehat{\boldsymbol{e}}_N\right]^\top$, $\boldsymbol{E}_3$ is an identity matrix.

In what follows, we aim to prove that if $\left|\widehat{e}_{i,j}\right| < p(t) = \sigma_{\min}\{\boldsymbol{L} + \boldsymbol{B}\}\rho(t)/\sqrt{3N}$, then $\|\boldsymbol{e}_i\| < \rho(t)$ holds, where $\sigma_{\min}\{\boldsymbol{L} + \boldsymbol{B}\} \leq \frac{\left(\frac{N-1}{N}\right)^{\frac{N-1}{2}}}{N^2+N-1}$ [42] is the minimum singular value of $\boldsymbol{L} + \boldsymbol{B}$.

Theorem 1: If auxiliary tracking error $\widehat{e}_i$ satisfies $\left|\widehat{e}_{i,j}\right| \leq p(t) = \sigma_{\min}\{\boldsymbol{L} + \boldsymbol{B}\}\rho(t)/\sqrt{3N}$, then it holds that $\|\boldsymbol{e}_i\| \leq \rho(t)$.

proof: According to the fact that graph $\bar{\mathcal{E}}$ is connected, it is able to verify that $\boldsymbol{L} + \boldsymbol{B}$ is positive definite [43]. Note that $\|X \otimes Y\| = \|X\|\,\|Y\|$, then, from (8), one can obtain that

$$\begin{aligned} \|\boldsymbol{e}(t)\| &= \left(\boldsymbol{E}_3 \otimes (\boldsymbol{L} + \boldsymbol{B})^{-1}\right)\widehat{e}(t) \\ &\leq \left\|(\boldsymbol{L} + \boldsymbol{B})^{-1}\right\| \|\boldsymbol{E}_3\| \left\|\widehat{e}(t)\right\| \\ &\leq \left\|\widehat{e}(t)\right\| / \sigma_{\min}\{\boldsymbol{L} + \boldsymbol{B}\} = \sqrt{\sum_{i=1}^{N} \widehat{e}_{i,j}^2} / \sigma_{\min}\{\boldsymbol{L} + \boldsymbol{B}\} \end{aligned} \tag{9}$$

Since $\left|\widehat{\boldsymbol{e}}_{i,j}\right| < p(t)$, then one has

$$\|\boldsymbol{e}(t)\| < \sqrt{3N}p(t)/\sigma_{\min}\{\boldsymbol{L} + \boldsymbol{B}\} = \rho(t) \tag{10}$$

By means of definition of $\boldsymbol{e}$, it follows that $\|\boldsymbol{e}_i\| \leq \|\boldsymbol{e}\| < \rho(t)$. This completes the proof.

B. CONTROLLER DESIGN

In what follows, based on the backstepping method, we are ready to design a low-complexity distributed controller such that the prescribed performance constraints $\|\boldsymbol{e}_i\| < \rho(t)$ and $\left|\varepsilon_{i,j}\right| < \rho(t)$ can be guaranteed, where $\boldsymbol{\varepsilon} = \left[\boldsymbol{\varepsilon}_1^T, \boldsymbol{\varepsilon}_2^T, \cdots, \boldsymbol{\varepsilon}_N^T\right]^T$ is the virtual tracking error, given by $\boldsymbol{\varepsilon} = \boldsymbol{v} - \boldsymbol{\alpha}$, $\boldsymbol{v} = \left[\boldsymbol{v}_1^T, \boldsymbol{v}_2^T, \cdots, \boldsymbol{v}_N^T\right]^T$, $\boldsymbol{\varepsilon}_i = \left[\varepsilon_{i,1}, \varepsilon_{i,2}, \varepsilon_{i,3}\right]^T$, $\boldsymbol{\alpha} = \left[\boldsymbol{\alpha}_1^T, \boldsymbol{\alpha}_2^T, \cdots, \boldsymbol{\alpha}_N^T\right]^T$ is the virtual control law, $\boldsymbol{\alpha}_i = \left[\alpha_{i,1}, \alpha_{i,2}, \alpha_{i,3}\right]^T$.

Let $\boldsymbol{s} = \left[\boldsymbol{s}_1^T, \boldsymbol{s}_2^T, \cdots, \boldsymbol{s}_N^T\right]^T$, $\boldsymbol{s}_i = \left[s_{i,1}, s_{i,2}, s_{i,3}\right]^T$, $s_{i,j} = \tanh^{-1}\left(\zeta_{i,j}\right)$, $\boldsymbol{z} = \left[\boldsymbol{z}_1^T, \boldsymbol{z}_2^T, \cdots, \boldsymbol{z}_N^T\right]^T$, $\boldsymbol{z}_i = \left[z_{i,1}, z_{i,2}, z_{i,3}\right]^T$, $z_{i,j} = \tanh^{-1}\left(\xi_{i,j}\right)$, $\boldsymbol{\zeta} = \left[\boldsymbol{\zeta}_1^T, \boldsymbol{\zeta}_2^T, \cdots, \boldsymbol{\zeta}_N^T\right]^T$, $\boldsymbol{\zeta}_i = \left[\zeta_{i,1}, \zeta_{i,2}, \zeta_{i,3}\right]^T$, $\zeta_{i,j} = \widehat{\boldsymbol{e}}_{i,j}/p$, $\xi = \left[\xi_1^T, \xi_2^T, \cdots, \xi_N^T\right]^T$, $\xi_i = \left[\xi_{i,1}, \xi_{i,2}, \xi_{i,3}\right]^T$, $\xi_{i,j} = \varepsilon_{i,j}/\rho(t)$, $\boldsymbol{\psi} = \left[\boldsymbol{\zeta}^\top, \xi^\top\right]^\top$, and $\mathcal{C} = \underbrace{(-1,1) \times \cdots \times (-1,1)}_{6N}$.

The boundedness of system signals is proven by the following two steps. We first demonstrate the boundedness of system signals in finite-time interval, ie., $\forall t \in [0, t_{\max})$. Subsequently, based on classical Theorem [44], the boundedness of system signals is established in infinite time interval, i.e., $\forall t \in [0, t_{\max})$. Through these two components, the overall bounded stability is rigorously proven.

Since $p(t)$, $\rho(t)$ and system (3) are bounded and smooth, and $\boldsymbol{\psi}(0) \in \mathcal{C}$ is guaranteed by the choice of $\widehat{\boldsymbol{e}}_{i,j}(0) < p(0)$ and $\|\boldsymbol{\varepsilon}_i(0)\| < \rho(0)$, it follows from Theorem 1 in [44] that there exists a unique and maximal solution $\boldsymbol{\psi}(t) \in \mathcal{C}$, $\forall t \in [0, t_{\max})$, where $t_{\max} < \infty$.

As proven in Theorem 1, to achieve $\|\boldsymbol{e}_i\| < \rho(t)$, we only need to ensure $\left|\widehat{\boldsymbol{e}}_{i,j}\right| < p(t)$. Towards this goal, we start by

defining the following candidate Lyapunov function

$$V_1 = \frac{1}{2}\boldsymbol{s}^T\boldsymbol{s} \tag{11}$$

Then the time derivative of V_1 yields

$$\begin{aligned}\dot{V}_1 &= \boldsymbol{s}^T\dot{\boldsymbol{s}}\\ &= p^{-1}\boldsymbol{s}^T\boldsymbol{G}\left(\left(\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3\right)\dot{e}(t)-\dot{p}\boldsymbol{\zeta}\right)\\ &= p^{-1}\boldsymbol{s}^T\boldsymbol{G}\left(\left(\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3\right)\left(\dot{\boldsymbol{x}}-\dot{\bar{x}}_d\right)-\dot{p}\boldsymbol{\zeta}\right)\end{aligned} \tag{12}$$

where $\boldsymbol{G} = diag\left\{\boldsymbol{G}_1,\cdots,\boldsymbol{G}_N\right\}$ and $\boldsymbol{G}_i = \text{diag}\left\{\left(1-\zeta_{i,j}^2\right)^{-1}\right\}$, $i = 1, 2, 3$. Then, it can be obtained that

$$\dot{V}_1 = p^{-1}\boldsymbol{s}^T\boldsymbol{G}\left(\left(\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3\right)\left(\boldsymbol{\varepsilon}+\boldsymbol{\alpha}-\dot{\bar{x}}_d\right)-\dot{p}\boldsymbol{\zeta}\right) \tag{13}$$

The virtual control law is designed as below

$$\boldsymbol{\alpha}_i = \kappa_1\boldsymbol{G}_i\boldsymbol{s}_i \tag{14}$$

where $\kappa_1 > 0$. By substituting (13) into (14), it yields

$$\begin{aligned}\dot{V}_1 &= p^{-1}\boldsymbol{s}^T\boldsymbol{G}\left(\left(\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3\right)\left(\boldsymbol{\varepsilon}-\kappa_1\boldsymbol{G}\boldsymbol{s}-\dot{\bar{x}}_d\right)-\dot{p}\boldsymbol{\zeta}\right)\\ &= -p^{-1}\kappa_1\boldsymbol{s}^T\boldsymbol{G}\left(\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3\right)\boldsymbol{G}\boldsymbol{s}\\ &\quad + p^{-1}\boldsymbol{s}^T\boldsymbol{G}\left(\left(\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3\right)\left(\varepsilon-\dot{\bar{x}}_d\right)-\dot{p}\boldsymbol{\zeta}\right)\end{aligned} \tag{15}$$

Since $\left(\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3\right)$ is positive matrix, then there exists a positive constant $\lambda_1 \geq \lambda_{\min}\left\{\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3\right\}$, where $\lambda_{\min}\{A\}$ is the minimum eigenvalue of matrix A. Then, (15) becomes

$$\begin{aligned}\dot{V}_1 &\leq -p^{-1}\kappa_1\lambda_1\left\|\boldsymbol{G}\boldsymbol{s}\right\|\\ &\quad + p^{-1}\boldsymbol{s}^T\boldsymbol{G}\left(\left(\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3\right)\left(\varepsilon-\dot{\bar{x}}_d\right)-\dot{p}\boldsymbol{\zeta}\right)\end{aligned} \tag{16}$$

Noting that $\boldsymbol{\psi}(t)\in\mathcal{C}, \forall t\in[0,t_{\max})$, it follows that $\boldsymbol{s}$, $\boldsymbol{\varepsilon}$, ζ and $\boldsymbol{\xi}$ are bounded $\forall t\in[0,t_{\max})$. Since p^{-1}, $\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3$ and $\dot{\bar{\boldsymbol{x}}}_d$ are bounded by the given assumption and structure, then we have

$$\begin{aligned}&\left\|p^{-1}\left(\left(\left(\boldsymbol{L}+\boldsymbol{B}\right)\otimes\boldsymbol{E}_3\right)\left(\boldsymbol{\varepsilon}-\dot{\bar{\boldsymbol{x}}}_d\right)-\dot{p}\boldsymbol{\zeta}\right)\right\|\\ &\qquad\leq\vartheta_1,\quad\forall t\in[0,t_{\max})\end{aligned} \tag{17}$$

In view of (16) and (17), one has

$$\begin{aligned}\dot{V}_1 &\leq -p^{-1}\kappa_1\lambda_1\left\|\boldsymbol{G}\boldsymbol{s}\right\|^2+\vartheta_1\left\|\boldsymbol{G}\boldsymbol{s}\right\|\\ &= -\left\|\boldsymbol{G}\boldsymbol{s}\right\|\left(\sigma_1\left\|\boldsymbol{G}\boldsymbol{s}\right\|-\vartheta_1\right)\end{aligned} \tag{18}$$

where $\sigma_1 = p^{-1}\kappa_1\lambda_1$. It is obvious that $\dot{V}_1 \leq 0$ if $\left\|\boldsymbol{G}\boldsymbol{s}\right\| \geq \vartheta_1/\sigma_1$, which implies that $\boldsymbol{G}\boldsymbol{s} = \boldsymbol{\beta}$ is bounded on $\forall t \in [0, t_{\max})$. Since $\zeta_{i,j}\in\mathcal{C}, \forall t\in[0,t_{\max})$, then one can obtain that $\left(1-\zeta_{i,j}^2\right)^{-1} > 0$ is bounded. According to the definition of $\boldsymbol{G}$ and the boundedness of $\boldsymbol{\beta}$, it is straightforward to obtain that $\left\|\boldsymbol{s}\right\| = \left\|\boldsymbol{G}^{-1}\right\|\left\|\boldsymbol{\beta}\right\|$ is bounded $\forall t\in[0,t_{\max})$. From (14), $\boldsymbol{\alpha}$ is bounded $\forall t\in[0,t_{\max})$.

Then, in order to achieve prescribed performance constraints $\left|\varepsilon_{i,j}\right| < \rho(t)$, we define the following candidate Lyapunov function

$$V_{2,i} = \frac{1}{2}z_i^\top z_i \tag{19}$$

Then the time derivative of $V_{2,i}$ yields

$$\begin{aligned}\dot{V}_{2,i} &= z_i^\top\dot{z}_i = \rho^{-1}z_i^\top\boldsymbol{H}_i\left(\dot{\varepsilon}_i-\dot{\rho}\xi_i\right)\\ &= \rho^{-1}z_i^\top\boldsymbol{H}_i\left(\dot{\boldsymbol{v}}_i-\dot{\boldsymbol{\alpha}}_i-\dot{\rho}\xi_i\right)\\ &= \rho^{-1}z_i^\top\boldsymbol{H}_i\left(\boldsymbol{C}_i\boldsymbol{v}_i+\boldsymbol{u}_i+\boldsymbol{d}_i-\dot{\boldsymbol{\alpha}}_i-\dot{\rho}\xi_i\right)\end{aligned} \tag{20}$$

where $\boldsymbol{H}_i = \text{diag}\left\{\left(1-\xi_{i,j}^2\right)^{-1}\right\}$, $i = 1, 2, 3$.

The control law is designed as below

$$\boldsymbol{u}_i = -\kappa_{2,i}\left(\left\|\boldsymbol{H}_iz_i\right\|^2+\mu_i\right)^{\frac{1}{2}}\boldsymbol{H}_iz_i \tag{21}$$

where $\kappa_1 > 0$. By substituting (21) into (20), it yields

$$\begin{aligned}\dot{V}_{2,i} &= -\rho^{-1}\kappa_{2,i}\left(\left\|\boldsymbol{H}_iz_i\right\|^2+\mu_i\right)^{\frac{1}{2}}z_i^\top\boldsymbol{H}\boldsymbol{H}_iz_i\\ &\quad+\rho^{-1}z_i^\top\boldsymbol{H}_i\left(\boldsymbol{C}_i\boldsymbol{v}_i+\boldsymbol{d}_i-\dot{\alpha}_i-\dot{\rho}\xi_i\right)\end{aligned} \tag{22}$$

Since $\boldsymbol{\psi}(t)\in\mathcal{C}, \forall t\in[0,t_{\max})$, then it is evident that $\boldsymbol{s}$, $\boldsymbol{\varepsilon}$, $\boldsymbol{\zeta}$ and $\boldsymbol{\xi}$ are bounded on $\forall t\in[0,t_{\max})$, which further implies that $\boldsymbol{\alpha}$ is bounded on $\forall t\in[0,t_{\max})$ by virtue of (14). In view of the boundedness of $\boldsymbol{e}$, $\bar{x}_d$, ε and $\boldsymbol{\alpha}$, we have that $\boldsymbol{x}$ and $\boldsymbol{v}$ are bounded. From $\dot{\boldsymbol{\alpha}}_i = \frac{\partial\boldsymbol{\alpha}_i}{\partial\boldsymbol{x}_i}\dot{\boldsymbol{x}}_i+\frac{\partial\boldsymbol{\alpha}_i}{\partial t}$, the fact that $\frac{\partial\boldsymbol{\alpha}_i}{\partial\boldsymbol{x}_i}$ and $\frac{\partial\boldsymbol{\alpha}_i}{\partial t}$ are smooth on $\forall t\in[0,t_{\max})$ by the definition and the boundedness of $\boldsymbol{x}$ and $\boldsymbol{v}$ on $\forall t\in[0,t_{\max})$, $\dot{\boldsymbol{\alpha}}_i$ is bounded on $\forall t\in[0,t_{\max})$. Noting that p^{-1}, C_i, $\boldsymbol{v}_i$, $\dot{\boldsymbol{\alpha}}_i$, $\dot{p}$ and ξ_i are bounded on $\forall t\in[0,t_{\max})$, there exists positive constant μ_i such that the following constraint holds

$$\begin{aligned}&\left\|\rho^{-1}z_i^\top\boldsymbol{H}_i\left(\boldsymbol{C}_i\boldsymbol{v}_i+\boldsymbol{d}_i-\dot{\boldsymbol{\alpha}}_i-\dot{\rho}\xi_i\right)\right\|\\ &\leq\mu_i\left\|\boldsymbol{H}_iz_i\right\|,\forall t\in[0,t_{\max})\end{aligned} \tag{23}$$

In view of (22), (23) and the fact that $\left\|\boldsymbol{H}_iz_i\right\|^2+\mu_i \geq \left\|\boldsymbol{H}_iz_i\right\|^2$, one can obtain that

$$\begin{aligned}\dot{V}_{2,i} &= -\rho^{-1}\kappa_{2,i}\left(\left\|\boldsymbol{H}_iz_i\right\|^2+\mu_i\right)^{\frac{1}{2}}\left\|\boldsymbol{H}_iz_i\right\|^2\\ &\quad+\mu_i\left\|\boldsymbol{H}_iz_i\right\| \leq -\left\|\boldsymbol{H}_iz_i\right\|\left(\rho^{-1}\kappa_{2,i}\left\|\boldsymbol{H}_iz_i\right\|-\mu_i\right)\end{aligned} \tag{24}$$

It is obvious that $\dot{V}_{2,i} \leq 0$ if $\left\|\boldsymbol{H}_iz_i\right\| \geq \mu_i/\left(\rho^{-1}\kappa_{2,i}\right)$, which implies that $\boldsymbol{H}_iz_i = \boldsymbol{\sigma}_i$ is bounded on $\forall t\in[0,t_{\max})$. Since $\boldsymbol{\psi}(t)\in\mathcal{C}, \forall t\in[0,t_{\max})$, then one can obtain that $\left(1-\xi_{i,j}^2\right)^{-1} > 0$ is bounded, which indicates that $\boldsymbol{H}_i$ is bounded on $\forall t\in[0,t_{\max})$. According to the boundedness of $\boldsymbol{H}_i$ and $\boldsymbol{\sigma}_i$, it is straightforward to obtain that $\left\|z_i\right\| = \left\|\boldsymbol{H}_i^{-1}\right\|\left\|\boldsymbol{\sigma}_i\right\|$ is bounded on $\forall t\in[0,t_{\max})$. From(21), $\boldsymbol{u}$ is bounded on $\forall t\in[0,t_{\max})$. Then, according to the definition of signals in closed-loop, all signals in closed-loop are bounded $\forall t\in[0,t_{\max})$.

C. STABILITY ANALYSIS

Theorem 2: Suppose that Assumptions 1-3 hold. Applying the virtual control law (14) and control law (21) into formation system (3), if $\boldsymbol{\psi}(0)\in\mathcal{C}$ holds, the following statements hold.

1) All signals are semi-globally ultimately uniformly bounded.

2) The fixed-time full-state error performance $\left\|\boldsymbol{e}_i\right\| < \rho(t)$ and $\left|\varepsilon_{i,j}\right| < \rho(t)$ can be guaranteed.

proof: We first prove that all signals of closed-loop system are semi-globally ultimately uniformly bounded $\forall t \in [0, t_{\max})$ with $t_{\max} = +\infty$. Since there exists a unique and maximal solution $\boldsymbol{\psi}(t) \in \mathcal{C}, \forall t \in [0, t_{\max})$, then, according to Theorem 2 in [42], two cases are possible: either $t_{\max} = +\infty$ or $\lim_{t \to t_{\max}-} \left[\|\boldsymbol{\psi}\| + \frac{1}{d_{\mathcal{S}}(\boldsymbol{\psi}, \mathcal{C})} \right] = \infty$. From (18) and (24), one can obtain that $\lim_{t \to t_{\max}-} \left[\|\boldsymbol{\psi}\| + \frac{1}{d_{\mathcal{S}}(\boldsymbol{\psi}, \mathcal{C})} \right] < \infty$.

Thus, we can conclude that all signals of closed-loop system are semi-globally ultimately uniformly bounded. Since $s_{i,j}$ and $z_{i,j}$ are bounded, the fixed-time prescribed performance constraints $\|\boldsymbol{e}_i\| < \rho(t)$ and $|\varepsilon_{i,j}| < \rho(t)$ can be guaranteed. This completes the proof.

Remark 1: This paper investigates a high-gain control method, which may produce high control gain. However, additional observers or adaptive laws are not required for addressing external disturbances and uncertainties. Meanwhile, the fixed-time full-state error performance for UAV formation can be guaranteed and the computation burden and communication burden are reduced.

Remark 2: This paper reduces computational complexity by eliminating the need for backstepping and its associated repeated differentiation steps. Unlike traditional backstepping controllers, our proposed fully distributed controller design avoids the iterative calculations and derivative computations inherent in that method. This simplification significantly reduces the computational burden on each UAV. The absence of a state observer further decreases complexity. Existing methods often rely on observers to estimate the states of neighboring UAVs, adding significant computational overhead. Our approach leverages only the readily available position information from neighbors, bypassing the need for this computationally expensive estimation process.

IV. SIMULATION EXAMPLE

Consider a UAV formation with one virtual leader and three followers, whose communication relationship graph is shown in FIGURE. 2. The relative position $\boldsymbol{\delta}_i$ between i-th follower and the virtual leader is given below

$$\begin{aligned} \boldsymbol{\delta}_1 &= \begin{bmatrix} 10 & 0 & 0 \end{bmatrix}^\top \\ \boldsymbol{\delta}_2 &= \begin{bmatrix} 0 & 10 & 0 \end{bmatrix}^\top \\ \boldsymbol{\delta}_3 &= \begin{bmatrix} -10 & 0 & 0 \end{bmatrix}^\top \end{aligned} \tag{25}$$

The initial state and model parameters are given below

$$\begin{aligned} &\boldsymbol{x}_0 = [0, 0, 10]^\top, \quad \boldsymbol{x}_1 = [20, 0, 5]^\top \\ &\boldsymbol{x}_2 = [0, 20, 7]^\top, \quad \boldsymbol{x}_3 = [-6, 0, 2]^\top \\ &\boldsymbol{v}_0 = [0, 0, 0]^\top, \quad \boldsymbol{v}_1 = [0.1, 0.1, 0.3]^\top \\ &\boldsymbol{v}_2 = [0.01, 0.2, -0.05]^\top, \quad \boldsymbol{v}_3 = [0.1, 0.1, 0.2]^\top \\ &\boldsymbol{C}_1 = \begin{bmatrix} 1 & & \\ & 2 & \\ & & 1 \end{bmatrix}, \quad \boldsymbol{C}_2 = \begin{bmatrix} 2 & & \\ & 1 & \\ & & 1 \end{bmatrix}, \boldsymbol{C}_3 = \begin{bmatrix} 3 & & \\ & 2 & \\ & & 1 \end{bmatrix} \\ &\boldsymbol{d}_1 = 0.1 \begin{bmatrix} \sin(0.1t) & \sin(0.3t) & \cos(0.1t) \end{bmatrix}^\top \\ &\boldsymbol{d}_2 = 0.2 \begin{bmatrix} \sin(0.1t) & \sin(0.3t) & \cos(0.1t) \end{bmatrix}^\top \\ &\boldsymbol{d}_3 = 0.3 \begin{bmatrix} \sin(0.1t) & \sin(0.3t) & \cos(0.1t) \end{bmatrix}^\top \end{aligned}$$

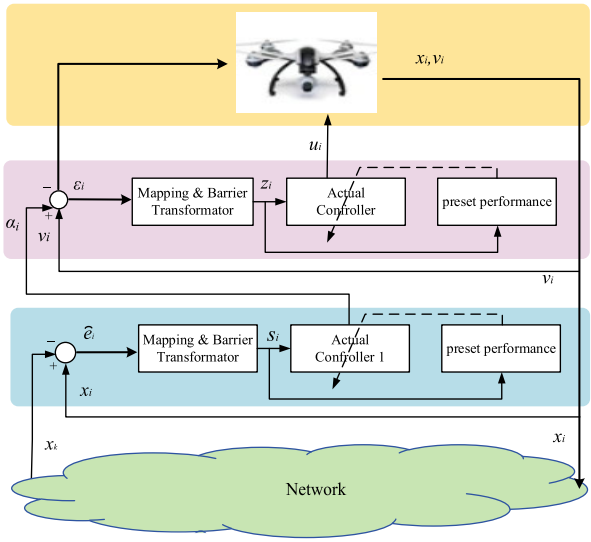

FIGURE 2. Architecture diagram of UAV formation control.

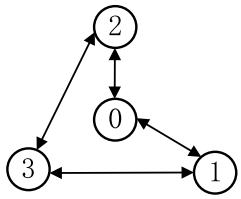

FIGURE 3. The communication graph.

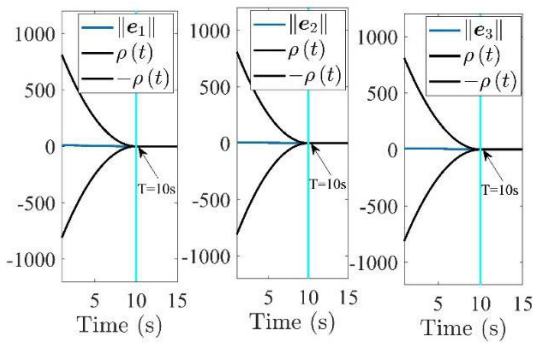

FIGURE 4. The formation tracking errors under fixed-time predefined performance.

The parameters of fixed-time prescribed performance function in (6) are chosen as $\rho_0 = 1000$, $\rho_\infty = 0.001$, $T = 10s$, With such design, initial value of formation tracking error $\boldsymbol{e}_i$ and virtual tracking error $\boldsymbol{\varepsilon}_i$ can satisfy $\|\boldsymbol{e}_i(0)\| < \rho(0)$ and $|\varepsilon_{i,j}(0)| < \rho(0)$, where $i = 1, 2, 3$ and $j = 1, 2, 3$. In addition, formation tracking error $\boldsymbol{e}_i$ and virtual tracking error $\boldsymbol{\varepsilon}_i$ can convergence into stability set $\|\boldsymbol{e}_i\| < \rho_\infty = 10^{-3}$ and $|\varepsilon_{i,j}| < \rho_\infty = 10^{-3}$ within fixed time $T = 10s$, respectively. The parameters of virtual controller (14) and controller (21) are set as $\kappa_{1,i} = 2, \kappa_{2,i} = 100$, respectively. Then, the simulation results are plotted in FIGURE. 4, FIGURE. 5, FIGURE. 6, FIGURE. 7. FIGURE. 8.

Figure 4 and Figure 5 illustrate that the formation tracking errors converge to predefined steady-state accuracy $\|\boldsymbol{e}_i\| < 10^{-3}$ within a fixed time $T = 10$ s. Figure 4 shows

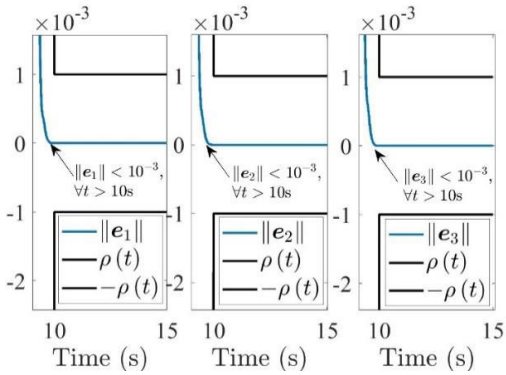

FIGURE 5. The partial magnification graph of formation tracking errors.

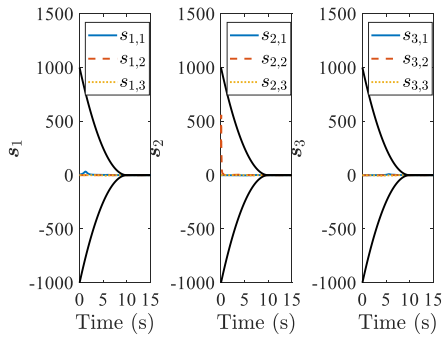

FIGURE 6. Virtual tracking error with fixed-time prescribed performance.

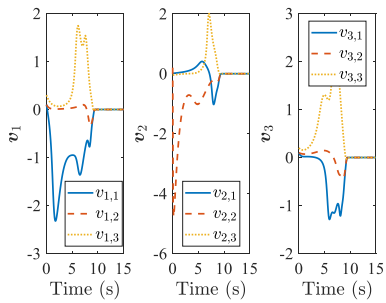

FIGURE 7. Velocity.

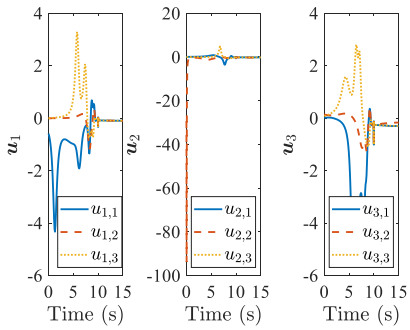

FIGURE 8. Control input.

the formation tracking errors under fixed-time predefined performance and Figure 5 is the partial magnification graph of formation tracking errors.

FIGURE. 6 shows the virtual velocity tracking errors of three UAVs over time, and these errors are confined to a preset performance boundary within a fixed time $T = 10s$.

FIGURE. 7 shows the speed variations of three UAVs. As three UAVs reach a steady value around 10 seconds, which aligns with the preset time. This indicates that the controller design can effectively control the movement of the UAVs.

V. CONCLUSION

In this paper, we have developed a fully distributed, low-complexity UAV formation controller with fixed-time full-state error performance. Under the proposed control scheme, only the position information of neighbors is required for communication, thereby effectively reducing the need for global information and decreasing the communication load, making it ideal for large-scale UAV formations. Unlike existing control methods that depending on adaptive parameters and complex laws, this approach is able to handle external disturbances without such requirements, simplifying the construction of controller and reducing computational demands to enhance real-time performance. Additionally, fixed-time full-state error performance in the presence of external disturbance and model uncertainty. The effectiveness of the proposed control method has been verified by conducting a numerical simulation. Designing a formation controller with practical physical constrains and fixed-time full-state error performance is an interesting topic for future investigation.

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